Assignment 4

1 Attention Exploration (14 points)

(a) i. The categorical distribution α puts almost all of its weight on α_j when the score $k_j^\top q$ is much larger than every other score $k_i^\top q$; equivalently, $\sum_{i \neq j} \exp(k_i^\top q - k_j^\top q) \ll 1$.

ii. Under this condition, $\alpha_j \approx 1$ and $\alpha_i \approx 0$ for every $i \neq j$. Therefore, $c = \sum_{i=1}^{n} \alpha_i v_i \approx v_j$.

(b) Let $\ell > 0$ be large and choose

$$q = \ell(k_a + k_b).$$

Since the keys are orthogonal unit vectors, $k_a^\top q = k_b^\top q = \ell$, whereas $k_i^\top q = 0$ for $i \notin \{a, b\}$. Hence

$$\alpha_a = \alpha_b = \frac{e^\ell}{2e^\ell + (n-2)} = \frac{1}{2 + (n-2)e^{-\ell}} \approx \frac{1}{2}$$

whenever $e^\ell \gg n - 2$. The remaining attention weights are negligible, so $c \approx \frac{1}{2}(v_a + v_b)$.

(c) i. To distinguish the covariance scale from the attention weights α_i, let ε denote the scalar called α in the question. Thus, $\Sigma_i = \varepsilon I$ with $\varepsilon \to 0$. Choose

$$q = \ell(\mu_a + \mu_b), \qquad e^\ell \gg n - 2.$$

Write $k_i = \mu_i + \delta_i$, where $\delta_i \sim \mathcal{N}(0, \varepsilon I)$. Since the means are orthogonal unit vectors,

$$k_a^\top q \approx k_b^\top q \approx \ell, \qquad k_i^\top q \approx 0 \quad (i \notin \{a, b\}).$$

Therefore, $\alpha_a \approx \alpha_b \approx \frac{1}{2}$ and $c \approx \frac{1}{2}(v_a + v_b)$. To make the conditions on ℓ precise, first note that the random score perturbation $\delta_i^\top q$ is a linear projection of a Gaussian vector and is therefore Gaussian. Its mean and variance are

$$\mathbb{E}[\delta_i^\top q] = 0, \qquad \mathrm{Var}(\delta_i^\top q) = q^\top (\varepsilon I) q = \varepsilon \|q\|^2 = 2\varepsilon\ell^2,$$

where $\|q\|^2 = \ell^2 \|\mu_a + \mu_b\|^2 = 2\ell^2$. Hence

$$\delta_i^\top q \sim \mathcal{N}(0, 2\varepsilon\ell^2),$$

whose standard deviation is $\sqrt{2}\ell\sqrt{\varepsilon}$. We may thus choose

$$\log(n-2) \ll \ell \ll \varepsilon^{-1/2}.$$

The lower bound implies $e^\ell \gg n - 2$, suppressing the total contribution of the irrelevant keys. The upper bound is equivalent to $\ell\sqrt{\varepsilon} \ll 1$, ensuring that the random score perturbations are small and that the scores of k_a and k_b remain approximately equal.

ii. Now $\Sigma_a = \varepsilon I + \frac{1}{2}\mu_a\mu_a^\top$. Ignoring the vanishing orthogonal noise, write $k_a \approx r\mu_a$, where the random magnitude

$$r = \mu_a^\top k_a \sim \mathcal{N}\left(1, \varepsilon + \frac{1}{2}\right).$$

Using the query from part (i), the two relevant scores satisfy

$$q^\top k_a \approx \ell r, \qquad q^\top k_b \approx \ell, \qquad \frac{\alpha_a}{\alpha_b} \approx e^{\ell(r-1)}.$$

Thus, the softmax exponentially amplifies fluctuations in r. When $r > 1$, c is close to v_a; when $r < 1$, c is close to v_b; only when r is very close to 1 is c close to $\frac{1}{2}(v_a + v_b)$. Consequently, individual samples produce outputs that vary substantially between vectors near v_a and v_b, so the variance of c is much larger than in part (i), even though its average over the symmetric fluctuations may remain near $\frac{1}{2}(v_a + v_b)$.

(d) i. Let $\ell > 0$ be large and choose two different queries

$$q_1 = \ell\mu_a, \qquad q_2 = \ell\mu_b.$$

When $\Sigma_i = \varepsilon I$ and $\varepsilon \to 0$, we have $k_i \approx \mu_i$. Thus, for q_1, the score of k_a is approximately ℓ, while all other scores are approximately zero; similarly, q_2 selects k_b. If $e^\ell \gg n - 1$ (and $\ell\sqrt{\varepsilon} \ll 1$), then

$$c_1 \approx v_a, \qquad c_2 \approx v_b, \qquad c = \frac{1}{2}(c_1 + c_2) \approx \frac{1}{2}(v_a + v_b).$$

ii. With $\Sigma_a = \varepsilon I + \frac{1}{2}\mu_a\mu_a^\top$, write $k_a \approx r\mu_a$, where $r \sim \mathcal{N}(1, \varepsilon + \frac{1}{2})$. The scores relevant to the first head are

$$q_1^\top k_a \approx \ell r, \qquad q_1^\top k_i \approx 0 \quad (i \neq a).$$

Ignoring the cases $q_1^\top k_a < 0$ as permitted, the first head still selects v_a whenever $\ell r \gg \log(n-1)$. Hence $c_1 \approx v_a$, although it has some residual variance when r is positive but close to zero. For the second head,

$$q_2^\top k_b \approx \ell, \qquad q_2^\top k_a \approx 0,$$

because $\mu_a^\top \mu_b = 0$. Therefore, $c_2 \approx v_b$ and has vanishingly small variance. Consequently,

$$c = \frac{1}{2}(c_1 + c_2) \approx \frac{1}{2}(v_a + v_b)$$

across samples. In particular, since c_2 is nearly deterministic,

$$\mathrm{Var}(c) \approx \frac{1}{4}\,\mathrm{Var}(c_1),$$

so the output is substantially more stable than the single-headed construction in part (c).

(e) Multi-headed attention assigns v_a and v_b to separate heads, so a change in the magnitude of k_a affects mainly the head responsible for v_a rather than changing the competition between v_a and v_b within one softmax. Averaging the head outputs therefore preserves both values more reliably and reduces the variance of the final output.

2 Position Embeddings Exploration (6 points)

(a) i. Since $X_{\text{perm}} = PX$, the query, key, and value matrices become

$$\begin{aligned} Q_{\text{perm}} &= X_{\text{perm}} W_Q = PXW_Q = PQ, \\ K_{\text{perm}} &= PK, \\ V_{\text{perm}} &= PV. \end{aligned}$$

Define $A = QK^\top/\sqrt{d}$. Then

$$\begin{aligned} H_{\text{perm}} &= \operatorname{softmax}\left(\frac{Q_{\text{perm}} K_{\text{perm}}^\top}{\sqrt{d}}\right) V_{\text{perm}} \\ &= \operatorname{softmax}(PAP^\top)PV \\ &= P \operatorname{softmax}(A) P^\top PV \\ &= P \operatorname{softmax}(A) V = PH, \end{aligned}$$

where we used $P^\top P = I_T$. A permutation leaves the all-ones vector unchanged, so $P\mathbf{1} = \mathbf{1}$. Consequently,

$$\begin{aligned} Z_{\text{perm}} &= \operatorname{ReLU}(H_{\text{perm}} W_1 + \mathbf{1}b_1) W_2 + \mathbf{1}b_2 \\ &= \operatorname{ReLU}\big(PHW_1 + P\mathbf{1}b_1\big) W_2 + P\mathbf{1}b_2 \\ &= \operatorname{ReLU}\big(P(HW_1 + \mathbf{1}b_1)\big) W_2 + P\mathbf{1}b_2 \\ &= P \operatorname{ReLU}(HW_1 + \mathbf{1}b_1) W_2 + P\mathbf{1}b_2 \\ &= P\,[\operatorname{ReLU}(HW_1 + \mathbf{1}b_1) W_2 + \mathbf{1}b_2] = PZ. \end{aligned}$$

Therefore, permuting the input rows by P permutes the output rows by the same matrix P.

ii. Although permutation equivariance is useful for unordered set inputs, it is problematic for text because word order carries syntactic and semantic information. Without positional information, permuting the input tokens merely permutes the output representations in the same way; after matching each output with its corresponding token, the representations are otherwise unchanged. Therefore, the Transformer cannot model order-dependent relationships, such as which noun is the subject and which is the object.

(b) i. Yes. After permuting the tokens, the position-encoded input is

$$X_{\text{pos,perm}} = PX + \Phi,$$

because Φ remains attached to the fixed sequence positions. In general,

$$PX + \Phi \neq P(X + \Phi) = PX + P\Phi.$$

Thus, the permutation-equivariance result from part (a) no longer applies. A token moved to a new position receives a different position embedding, allowing the Transformer to model word order.

ii. No. Suppose two integer positions t and t' had identical position embeddings. Taking $i = 0$ gives

$$\sin t = \sin t', \qquad \cos t = \cos t'.$$

Hence $t - t' = 2\pi m$ for some integer m. Since $t - t'$ is an integer, this is possible only when $m = 0$, so $t = t'$. Therefore, two distinct positions cannot have identical position embeddings.

3 Pretrained Transformer models and knowledge access (35 points)

(d) Finetune w/o pretraining: Correct: 11.0 out of 500.0: 2.1999999999999997%.

London baseline: Correct: 25.0 out of 500.0: 5.0%.

(f) Finetune w/ pretraining: Correct: 128.0 out of 500.0: 25.6%.

(g) i. For $k = 1, \ldots, d/2$, represent the k-th pair of features as the complex number

$$z_{t,k} = x_t^{(2k-1)} + \mathrm{i}x_t^{(2k)}, \qquad \theta_k = 10000^{-2(k-1)/d}.$$

The k-th element of the element-wise product in Equation 4 is

$$u_{t,k} = \big(\cos(t\theta_k) + \mathrm{i}\sin(t\theta_k)\big)\big(x_t^{(2k-1)} + \mathrm{i}x_t^{(2k)}\big).$$

Expanding this product and using $\mathrm{i}^2 = -1$ gives

$$\begin{aligned} u_{t,k} &= x_t^{(2k-1)}\cos(t\theta_k) + \mathrm{i}x_t^{(2k-1)}\sin(t\theta_k) \\ &\quad + \mathrm{i}x_t^{(2k)}\cos(t\theta_k) - x_t^{(2k)}\sin(t\theta_k) \\ &= \big[x_t^{(2k-1)}\cos(t\theta_k) - x_t^{(2k)}\sin(t\theta_k)\big] \\ &\quad + \mathrm{i}\big[x_t^{(2k-1)}\sin(t\theta_k) + x_t^{(2k)}\cos(t\theta_k)\big]. \end{aligned}$$

Therefore, the real and imaginary parts of $u_{t,k}$ satisfy

$$\begin{bmatrix} \mathrm{Re}(u_{t,k}) \\ \mathrm{Im}(u_{t,k}) \end{bmatrix} = \begin{bmatrix} \cos(t\theta_k) & -\sin(t\theta_k) \\ \sin(t\theta_k) & \cos(t\theta_k) \end{bmatrix} \begin{bmatrix} x_t^{(2k-1)} \\ x_t^{(2k)} \end{bmatrix}.$$

This is exactly the k-th 2×2 rotation block in Equation 3. Stacking the real and imaginary parts for all k in their original order gives

$$\mathrm{RoPE}(x_t, t) = \begin{bmatrix} \mathrm{Re}(u_{t,1}) \\ \mathrm{Im}(u_{t,1}) \\ \vdots \\ \mathrm{Re}(u_{t,d/2}) \\ \mathrm{Im}(u_{t,d/2}) \end{bmatrix},$$

which is the vector produced by the block-diagonal matrix in Equation 3. Thus, Equation 4 computes all two-dimensional rotations simultaneously by element-wise complex multiplication; in code, this requires reshaping the real vector into pairs, converting the pairs to complex numbers, applying the multiplication, converting back to real numbers, and flattening the pairs.

ii. For a two-dimensional vector represented by a complex number, the RoPE operation is

$$\mathrm{RoPE}(z, t) = e^{\mathrm{i}t\theta} z,$$

and the corresponding real dot product is

$$\langle z_1, z_2 \rangle = \mathrm{Re}(z_1 \overline{z_2}),$$

where the bar denotes complex conjugation. Therefore,

$$
\begin{aligned}
\langle \mathrm{RoPE}(z_1, t_1), \mathrm{RoPE}(z_2, t_2)\rangle &= \mathrm{Re}\left[\left(e^{\mathrm{i}t_1\theta} z_1\right)\overline{\left(e^{\mathrm{i}t_2\theta} z_2\right)}\right] \\
&= \mathrm{Re}\left[e^{\mathrm{i}t_1\theta} z_1 e^{-\mathrm{i}t_2\theta}\overline{z_2}\right] \\
&= \mathrm{Re}\left[e^{\mathrm{i}(t_1-t_2)\theta} z_1\overline{z_2}\right].
\end{aligned}
$$

On the other hand, since $\mathrm{RoPE}(z_2, 0) = e^0 z_2 = z_2$, we have

$$
\begin{aligned}
\langle \mathrm{RoPE}(z_1, t_1 - t_2), \mathrm{RoPE}(z_2, 0)\rangle &= \mathrm{Re}\left[\left(e^{\mathrm{i}(t_1-t_2)\theta} z_1\right)\overline{z_2}\right] \\
&= \mathrm{Re}\left[e^{\mathrm{i}(t_1-t_2)\theta} z_1\overline{z_2}\right].
\end{aligned}
$$

The two expressions are identical. Hence,

$$
\langle \mathrm{RoPE}(z_1, t_1), \mathrm{RoPE}(z_2, t_2)\rangle = \langle \mathrm{RoPE}(z_1, t_1 - t_2), \mathrm{RoPE}(z_2, 0)\rangle,
$$

showing that the dot product depends on the positions only through their relative displacement $t_1 - t_2$.

iii. Correct: 154.0 out of 500.0: 30.8%.

4 Considerations in pretrained knowledge (5 points)

(a) Pretraining on Wikipedia exposes the model to associations between people's names and their birthplaces, together with broader world knowledge. Fine-tuning then teaches the model to use these associations in the birthplace-prediction format. Because the fine-tuning dataset is small, a model trained from scratch cannot learn a sufficiently general mapping from names to places and therefore performs poorly on unseen names.

(b) 1. The model may produce a plausible but incorrect answer that a user cannot distinguish from a retrieved fact, causing the user to act on false information. For example, a medical assistant might invent a diagnosis or dosage, leading to an unsafe treatment decision.

2. The absence of reliable provenance or uncertainty estimates makes errors difficult to audit, correct, and assign responsibility for. For example, a legal assistant might cite an invented court decision, while giving the user no source that could be checked.

(c) For a person absent from both pretraining and fine-tuning, the model may rely on surface cues in the name, such as its spelling, formatting, or resemblance to names associated with particular regions, and generate the most likely birthplace. This is a heuristic guess rather than retrieved knowledge. Such behavior can reinforce stereotypes and lead to discriminatory misclassification; for example, an automated screening system might infer a person's nationality or birthplace from their surname and use that inference in an employment or immigration decision.